

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Senior AI and Data Engineering professional with **6+ years of experience** designing and delivering production-grade AI, data, and distributed software systems, including more than two years working with Generative AI, LLMs, autonomous agents, and enterprise agentic workflows.
- Hands-on experience architecting **multi-agent AI systems** using LangChain, LangGraph, LlamaIndex, AutoGen, Google ADK concepts, agent-to-agent (A2A) communication, task delegation hierarchies, Model Context Protocol (MCP), and tool/function calling.
- Strong expertise building **RAG systems, semantic search, ONNX-based embedding workflows, vector databases, agent memory, context management, stateful execution, prompt engineering, and token optimization** for scalable context-aware AI applications.
- Experienced across the full AI lifecycle including **feature engineering, statistical modeling, propensity/classification/forecasting models, training, validation, deployment, LLMOps, production monitoring, model drift detection, and continuous optimization**.
- Strong cloud and data engineering background with **Python, SQL, BigQuery, GCP Cloud Run, Vertex AI, Amazon Bedrock, Azure, REST APIs, distributed pipelines, and large-scale data extraction and transformation**.
- Implemented **AI governance, security guardrails, permission boundaries, agent identity controls, observability, OpenTelemetry-style tracing, evaluation pipelines, model validation, cost/token monitoring, and production reliability controls**.
- Demonstrated technical leadership through **AI strategy, stakeholder communication, business-impact analysis, mentoring, cross-functional collaboration, executive presentations, and translating probabilistic AI outputs into clear operational recommendations and measurable outcomes**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & AI Engineering:** Python, SQL, PySpark, Java, FastAPI, REST APIs, Statistical Modeling, Feature Engineering, Prompt Engineering, Token Management, Large-Scale Data Extraction.
- **Agentic AI & Multi-Agent Systems:** LangChain, LangGraph, LlamaIndex, AutoGen, Google ADK, Multi-Agent Orchestration, A2A Protocols, MCP, Task Delegation, Tool Calling, Agent Planning, Reasoning.
- **RAG & Agent Memory:** RAG, ONNX Embeddings, Semantic Search, Pinecone, Weaviate, pgvector, Short-Term Memory, Long-Term Memory, Context Window Optimization, Stateful Workflow Execution.
- **Cloud & Data Platforms:** GCP, BigQuery, Cloud Run, Vertex AI, Amazon Bedrock, AWS, Azure, Palantir Foundry/AIP, Data Pipelines, Distributed Processing, Cloud-Native Architecture.
- **ML & Commercial Analytics:** Propensity Modeling, Classification, Forecasting, Churn Prediction, Customer Lifetime Value (CLV), Generative AI-Augmented Analytics, Model Validation, Statistical Analysis, AI Decision Systems.
- **LLMOps & Observability:** OpenTelemetry, Arize Phoenix, Galileo Concepts, Model Drift Detection, Accuracy Degradation Monitoring, Agent Output Quality, Evaluation Pipelines, LLMOps, Logging, Tracing, Metrics, Alerting.
- **Governance & Delivery:** AI Governance, Security Guardrails, Compliance, Permission Boundaries, Agent Identity Tokens, Financial ROI Modeling, CI/CD, Git, Docker, Terraform, Executive Communication, Technical Leadership.

Software Engineer AI
PCMI LLC (Full Time)

Oct 2024 – Present
Dallas, TX, USA

- Architected and deployed production **multi-agent AI systems** using Python, Palantir Foundry/AIP, LangGraph, LangChain, LLMs, and cloud services, enabling autonomous agents to reason, plan, delegate tasks, use enterprise tools, and execute complex insurance and financial workflows.
- Designed **A2A communication and task delegation patterns** for cooperative agents and integrated enterprise capabilities through MCP, REST APIs, function calling, and structured tool interfaces, enabling scalable orchestration across multiple business domains.
- Built production-grade **RAG pipelines** using ONNX-compatible embeddings, semantic search, vector databases, metadata filtering, retrieval strategies, and context-aware prompting to generate accurate and personalized outputs from governed enterprise data.
- Implemented robust **agent memory and state architecture** using LangGraph-based workflows, short-term context, persistent long-term memory, context-window optimization, checkpoints, retries, and stateful multi-step execution for reliable autonomous processes.
- Owned the full AI lifecycle including **feature engineering, model integration, deployment, evaluation, monitoring, token/cost optimization, model-output validation, and production support**, while building propensity, classification, and forecasting capabilities for downstream agent decisions.
- Established enterprise AI controls through **security guardrails, RBAC, permission boundaries, short-lived identity concepts, audit logging, safety constraints, OpenTelemetry-style observability, drift detection, and agent-quality monitoring**.
- Presented AI findings and operational strategies to product, engineering, and business stakeholders, translating complex agent behavior into **business outcomes, ROI considerations, implementation roadmaps, and measurable improvements**, while mentoring engineers on agentic architecture and LLMops practices.

Environments: Python, SQL, Palantir Foundry/AIP, LangGraph, LangChain, LlamaIndex, Multi-Agent Systems, A2A, MCP, RAG, ONNX Embeddings, Pinecone/pgvector Concepts, Semantic Search, Agent Memory, AWS, Azure, Vertex AI Concepts, Bedrock, OpenTelemetry, LLMops, CI/CD, Docker, Terraform.

Software Engineer
Client: CVS

Dec 2023 – Sept 2024
Dallas, TX, USA

- Designed and developed production **AI-enabled healthcare and analytical systems** using Python, FastAPI, AWS, SQL, LLMs, and distributed services, delivering scalable decision-support capabilities across regulated enterprise workflows.
- Built agentic applications using **LangChain/LangGraph concepts, RAG, tool calling, structured prompting, state management, and multi-step orchestration**, allowing AI services to reason over contextual healthcare data and invoke enterprise functions.
- Developed retrieval workflows using **embeddings, semantic search, vector storage, metadata-aware retrieval, context optimization, and structured validation**, improving relevance and reliability of LLM-generated outputs.
- Engineered large-scale data and AI pipelines using **Python, SQL, AWS services, distributed processing, APIs, and event-driven patterns** for feature preparation, model inputs, inference outputs, analytics, and downstream consumption.
- Implemented AI monitoring and production controls with **CloudWatch, Splunk, structured logging, metrics, evaluation pipelines, latency monitoring, output-quality validation, drift detection concepts, and production diagnostics**.
- Applied secure deployment practices using **OAuth2/OIDC, JWT, IAM, RBAC, Docker, CI/CD, cloud security controls, and audit-ready logging**, supporting reliable operation of AI capabilities in a regulated healthcare environment.
- Collaborated with engineers, data scientists, architects, and business stakeholders to **interpret AI outputs, communicate model behavior, evaluate business impact, optimize decision workflows, and translate technical findings into clear operational recommendations**.

Environments: Python, SQL, FastAPI, AWS, LLMs, RAG, LangChain, LangGraph Concepts, Embeddings, Vector Databases, Semantic Search, Agent State, REST APIs, Event-Driven Architecture, OAuth2/OIDC, JWT, IAM, CloudWatch, Splunk, LLM Evaluation, Docker, CI/CD.

Software Engineer
Infor

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise **AI, analytics, and backend solutions** using Python, SQL, FastAPI, Flask, PySpark, and distributed-system architectures, supporting operational decision-making across complex business applications.
- Built analytical models and processing workflows for **classification, forecasting, feature engineering, statistical analysis, and predictive insights**, exposing outputs through reusable APIs and downstream business applications.
- Engineered scalable data pipelines using **Python, PySpark, SQL, REST APIs, batch processing, and event-driven patterns** for large-scale data extraction, transformation, aggregation, and model-feature generation.
- Developed service-orchestration patterns supporting **state management, asynchronous execution, retries, exception handling, API integrations, and reusable business logic**, establishing foundations for intelligent workflow automation.
- Improved model and service reliability through **structured logging, monitoring, performance analysis, data-quality checks, root-cause investigation, and production diagnostics** across distributed application components.
- Applied modern engineering practices including **Git, code reviews, Jenkins, CI/CD, Docker, technical documentation, API design, and production deployment**, improving maintainability and release consistency.
- Partnered with product teams, architects, analysts, and business stakeholders to **translate complex analytical outputs into actionable recommendations, explain technical trade-offs, and support data-driven operational strategies**.

Environments: Python, SQL, FastAPI, Flask, PySpark, Apache Spark, REST APIs, Microservices, Classification, Forecasting, Feature Engineering, Statistical Analysis, Distributed Processing, Git, Jenkins, CI/CD, Docker, Logging, Monitoring, Production Support.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data-driven backend and analytics applications** using Python, SQL, REST APIs, PostgreSQL, MySQL, and distributed application technologies to support operational decision-making across client environments.
- Built reusable Python components for **data extraction, transformation, feature preparation, statistical processing, API integration, workflow execution, and database interaction** across high-volume enterprise workloads.
- Developed analytical processing workflows using **Python, SQL, structured datasets, aggregation logic, trend analysis, and predictive modeling concepts** to surface actionable patterns from operational information.
- Implemented distributed integration patterns using **REST APIs, asynchronous workflows, event-driven concepts, state handling, error recovery, and relational data services**, improving reliability across multi-system processes.
- Applied operational controls including **logging, monitoring, exception handling, performance diagnostics, root-cause analysis, and data validation** to maintain reliable execution of production applications and data workflows.
- Supported production delivery through **Git, CI/CD, Docker, Linux, version control, deployment automation, documentation, and secure integration patterns** across distributed enterprise environments.
- Worked with architects, engineers, analysts, and business stakeholders to **interpret data findings, explain technical behavior, communicate risks and dependencies, and translate analytical insights into practical business recommendations**.

Environments: Python, SQL, PostgreSQL, MySQL, REST APIs, Data Processing, Feature Engineering Concepts, Statistical Analysis, Predictive Analytics Concepts, Event-Driven Architecture, Git, CI/CD, Docker, Linux, Logging, Monitoring, Root-Cause Analysis, Production Support.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.